

# Molecular Insights into Lithium-Ion Coordination and Morphology in Carbonate Polymer Electrolytes

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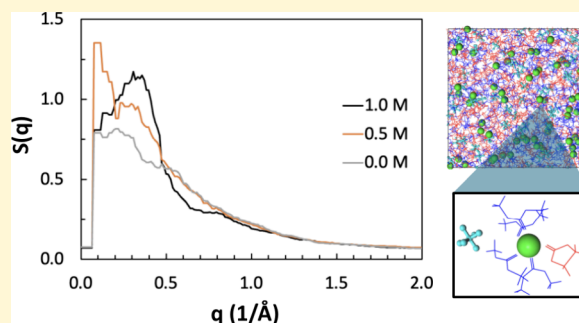


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**ABSTRACT:** Research of solid-state polymer electrolytes (SPEs) has accelerated due to their promise to address critical barriers in lithium-ion and lithium-metal battery commercialization through their superior thermal stability, reduced flammability, and mitigation of dendrite formation. In this study, we employ all-atom molecular dynamics simulations to investigate the  $\text{Li}^+$  solvation structure, ion diffusion, and phase morphology in mixtures of ethylene carbonate (EC) and dimethyl carbonate (DMC) across various salt concentrations. Our findings indicate that low salt concentrations diminish ionic interactions and enhance ion mobility, whereas elevated salt levels facilitate ion clustering and reduce ion mobility. Based on these findings, solid polymer electrolytes were designed using EC and DMC moieties. Polymers incorporating DMC exhibit greater backbone flexibility and lower glass transition temperatures resulting in an ion transport enhancement. The study also examines mixed-branch copolymer systems and polymer blend systems as alternative approaches for tuning mechanical and ionic transport properties. Both direct copolymerization and physical blending of single-branch polymers allow fine-tuning of mechanical and electrochemical properties. Notably, at elevated salt concentrations,  $\text{Li}^+$  ions act as compatibilizers that reduce phase separation. These findings contribute to a fundamental understanding of the relationships among the polymer structure, salt concentration, and ion transport in carbonate-based polymer electrolytes.



## 1. INTRODUCTION

The transition to sustainable energy storage systems necessitates developing electrochemical energy storage technologies that are both efficient and safe. Lithium-ion batteries (LIBs) have been central to this transformation and have proven critical to applications from portable electronics to electric vehicles.<sup>1–5</sup> However, conventional LIBs employing liquid electrolytes face critical challenges such as leakage, flammability, and chemical instability, which raises great safety concerns and limits their applicability.

Solid-state polymer electrolytes (SPEs) are a promising alternative for addressing these shortcomings while offering additional potential advantages.<sup>6–8</sup> SPEs tend to be mechanically robust and flexible materials that integrate the roles of both separator and electrolyte. Further, their nonvolatile nature eliminates leakage concerns, while at the same time their structural properties suppress dendrite formation, which is a critical concern preventing commercialization of lithium metal batteries.

Among various SPEs, poly(ethylene oxide) (PEO) has been extensively studied due to its capability to facilitate  $\text{Li}^+$  ion migration by coordinating the ions with ether oxygens along the chain.<sup>9–16</sup> Furthermore, its low glass transition temperature ( $T_g$ ) of approximately  $-50\text{ }^\circ\text{C}$ ,<sup>17</sup> primarily attributed to its high flexibility, can be promising for ion transport. However, it has been noted that, despite their low  $T_g$ , the formation of

semicrystalline phases at ambient conditions severely restricts the mobility of  $\text{Li}^+$  ions. As a result, their ionic conductivity at ambient temperatures is several orders of magnitude lower than that of liquid electrolytes,<sup>9</sup> and thereby, the high operating temperature condition required to achieve sufficient ionic conductivity in PEO-based SPEs raises other concerns about thermal stability and compatibility with other battery components. Additionally, due to the strong interaction between the ethylene oxide (EO) groups and  $\text{Li}^+$  ion, which forms a high coordination of EO with respect to  $\text{Li}^+$  ions, the migration of  $\text{Li}^+$  ion at room temperature is substantially limited. Although this could be addressed by reducing the concentration of the EO groups along the backbone, it would cause the incomplete dissociation of the Li salts, which is already suboptimal due to the low dielectric constant ( $\sim 8$ ) of PEO.

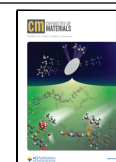
In this context, therefore, research efforts have focused on developing carbonate-based polymer electrolytes, particularly,

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aliphatic polycarbonates such as poly(trimethylene carbonate) (PTMC) and poly(ethylene carbonate) (PEC), due to its high dielectric constants and flexibility, which are particularly promising as they facilitate the dissociation of lithium salts and improve ion mobility.<sup>18–20</sup> However, it has been noted that the carbonate-based SPEs also face challenges such as high  $T_g$  in some formulations and suboptimal mechanical stability, requiring further optimization.<sup>7</sup>

Recent studies<sup>21,22</sup> have demonstrated that using copolymerization and polymer blend can provide unique advantages of ether- and carbonate-based polymer systems: for instance, the incorporation of high dielectric carbonate groups alongside flexible ether segments imparts the salt dissociation and the ion mobility to the system simultaneously. This approach can achieve a balance among ionic conductivity, mechanical stability, and electrochemical compatibility by synergizing the complementary properties of these functional groups.

While substantial progress has been made in the study of SPEs, the fundamental understanding of ionic transport mechanisms remains veiled, particularly for carbonate-based systems. The interactions among cations, anions, and polymer chains determine the structures and dynamics over multiple characteristic time scales, which complicates experimental decoupling and characterization. This lack of mechanistic understanding impedes the systematic optimization of these materials and the design of improved systems with desirable conductivity. In this context, computational modeling and simulation methods provide a framework to address the challenges by offering molecular-level insights into the relationships among polymer morphology, ionic coordination, clustering, and transport properties.

In this work, we use all-atom molecular dynamics (MD) simulations to investigate liquid-state carbonate systems and solid-state polymerized carbonate systems. Our simulation study reveals several key findings: (1) the solvation environment of  $Li^+$  ions shows unexpected concentration dependence, where  $Li^+$  solvation by dimethyl carbonate (DMC) is more dominant at lower concentrations while ethylene carbonate (EC) becomes competitive at higher concentrations (solvents depicted in Figure S2); (2) the polymer architecture fundamentally alters this solvation behavior compared to the low molecular weight counterpart in liquid systems due to the backbone constraints limiting the conformational degree of freedom typically available to DMC; (3) the  $Li^+$  concentration plays a dual role in polymer electrolytes - at moderate concentrations (0.5M), it enhances phase separation between EC and DMC domains, while at higher concentrations (1.0M), it acts as a compatibilizer, creating a more uniform morphology; (4) while higher  $Li^+$  concentrations promote morphological stability, they paradoxically lead to reduced ion mobility due to strong ion-polymer interactions and reduced chain dynamics.

The findings indicate that effective electrolyte performance necessitates balanced phase behavior, solvation structures, and polymer mobility, thereby establishing a foundation for comprehending ionic transport in carbonate-based solid polymer electrolytes (SPEs). The detailed analyses of solvation structure, ion clustering, and polymer morphology provide design insights for improving conductivity in SPEs through targeted structural modifications. We hope this aids the development of next-generation electrochemical energy storage technologies.

## 2. MODELS AND SIMULATION METHODS

**2.1. Model Construction and MD Simulation Conditions.** A Monte Carlo algorithm was used to initialize polymer chains in a simulation box, with a starting density of 0.2 g/cm<sup>3</sup>. The system was gradually heated to the target temperature of 298.15 K over a 1 ns period under constant volume and temperature (NVT) conditions to initiate system relaxation. Subsequently, constant pressure and temperature (NPT) simulations were performed to allow the system to adjust its volume naturally and achieve a denser state at ambient temperature and pressure conditions (298.15 K, 1 atm). To further obtain the equilibrium state of the system, a sequence of heating and cooling cycles under NPT conditions was applied. Finally, NPT simulations using the Nosé–Hoover thermostat were performed until the systems reached the equilibrium state. The analysis was subsequently implemented based on the data from these equilibrated states, typically over the last 40 ns of the simulations. Each polymer electrolyte simulation was conducted in a cubic periodic cell containing 44 polymer chains (degree of polymerization = 10) along with the appropriate number of  $LiPF_6$  ion pairs at each salt concentration (Table S2). The cells equilibrated under NPT (298.15 K, 1 atm) to edge lengths of 4.20–4.85 nm (Table S3). Although these box parameters are typical of other studies in the field,<sup>23,24</sup> larger simulations ( $\geq 8$  nm) could further confirm the full convergence of transport coefficients and phase morphology.

All MD simulations were performed in the LAMMPS (large-scale atomic/molecular massively parallel simulator) software package.<sup>25</sup> The Dreiding force field<sup>26</sup> described bonded and nonbonded interactions, except for  $Li^+$  ion interaction with solvent molecules, which were parametrized based on density functional theory (DFT) calculations.  $Li^+$  ion parameters were obtained from Jensen and Jorgensen.<sup>27</sup> Quantum mechanical simulations were performed using the DMol<sup>3</sup> module,<sup>28</sup> which employs a numerical basis set<sup>29</sup> for the description of electron density. The exchange–correlation was described using the Perdew–Burke–Ernzerhof (PBE)<sup>30,31</sup> functional under the generalized gradient approximation (GGA). The electronic basis set applied was the DNP 4.4 (Double Numerical plus Polarization basis set).<sup>32–37</sup> Charge analysis was performed using the Mulliken scheme.<sup>38–41</sup>

Nonbonded force field parameters were developed by fitting a morse potential to the DFT calculations for  $Li^+$  ion interactions with other electrolyte components:

$$E = D_0[e^{-2\alpha(r-r_0)} - 2e^{-\alpha(r-r_0)}] \quad (1)$$

where  $D_0$ ,  $\alpha$ ,  $r_0$ , and  $r$  correspond to the strength of the dispersion energy (or well depth), a parameter controlling the well width, the equilibrium distance between the two particles (where the energy is minimized), and the distance between the two particles (Table S1).

The Lennard-Jones 12–6 potential, which is generally more computationally efficient, is implemented for all other nonbonded interactions, including organic–organic interactions:

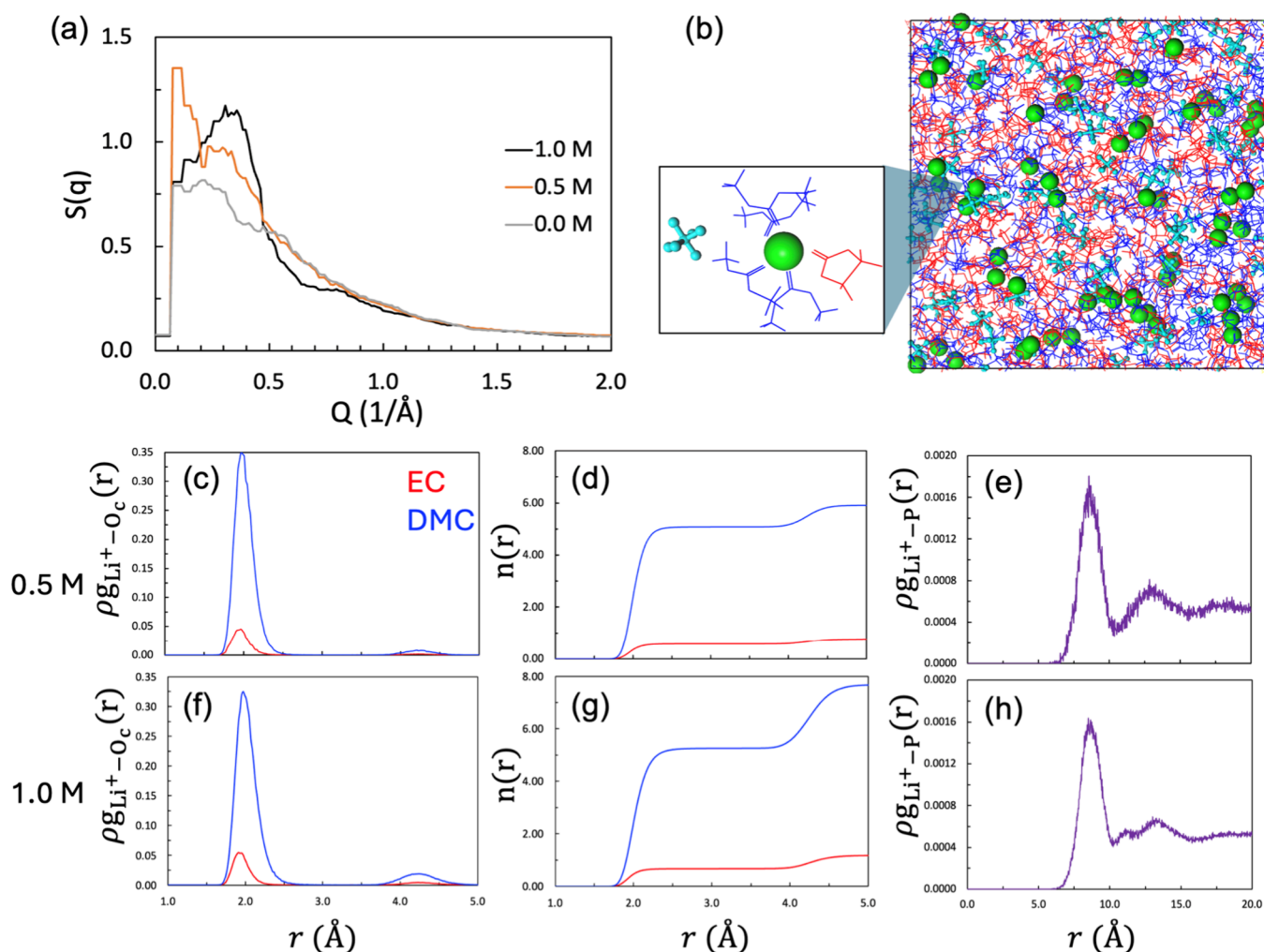
$$E = 4\epsilon \left[ \left( \frac{\sigma}{r} \right)^{12} - \left( \frac{\sigma}{r} \right)^6 \right] \quad (2)$$

where  $\epsilon$ ,  $\sigma$ , and  $r$  correspond to the strength of the dispersion energy.

**2.2. Structure Factor Analysis.** In order to characterize the extent of nanophase segregation between the EC and DMC components in the solvent systems, we employ the structure factor,  $S(q)$ ,<sup>42</sup> which provides insight into the spatial distribution of different components within the solvent blend (akin to small-angle scattering experiments). The structure factor,  $S(q)$ , is defined as follows:

$$S(q) = \frac{1}{V} \left\langle \sum_i \sum_j e^{iq \cdot r_{ij}} \times (\xi_i^e \xi_j^e - \langle \xi^2 \rangle) \right\rangle \quad (3)$$

where  $\xi_i$  and  $\xi_j$  represent local density contrasts at sites  $i$  and  $j$ , the scattering vector  $q$  corresponds to the inverse of a characteristic length scale,  $r_{ij}$  defines the vector separating  $i$  and  $j$ , and  $V$  denotes the system volume. In this formulation,  $\langle \dots \rangle$  corresponds to an ensemble average over thermal fluctuations. The term  $e^{iq \cdot r_{ij}}$  accounts the phase factor



**Figure 1.** (a) Structure factor of liquid EC/DMC electrolyte with 0.0, 0.5, and 1.0 M concentration of  $\text{LiPF}_6$ . (b) Depiction of equilibrated morphology and  $\text{Li}^+$  solvation shell in the liquid blend with 1.0 M concentration of  $\text{LiPF}_6$ . Systems are modeled at 298 K. (c, e) Radial distribution functions (RDFs) illustrating the spatial correlation between  $\text{Li}^+$ -ions and carbonyl oxygen atoms in ethylene carbonate (EC) and dimethyl carbonate (DMC), and RDFs depicting  $\text{Li}^+$ -ion proximity to phosphorus (P) in  $\text{PF}_6^-$  anions at a concentration of 0.5 M  $\text{LiPF}_6$ . (d) The local coordination numbers for EC and DMC surrounding  $\text{Li}^+$ -ions are presented. (f–h) Corresponding RDFs and local coordination numbers at an increased concentration of 1.0 M  $\text{LiPF}_6$ .

associated with the wave nature of scattering. In the context of scattering experiments, small-angle X-ray scattering (SAXS) captures variations in electron density, whereas small-angle neutron scattering (SANS) is sensitive to deuterium density differences. However, within our model, the structure factor is derived from an artificial density contrast.  $\Phi_i^A$  is a characteristic function for the local density at site  $i$  for some generic constituent  $A$ , such that  $\Phi_i^A = 1$  if site  $i$  is occupied by  $A$  – and  $\Phi_i^A = 0$  in the contrary case. The local density functions are related to the local density contrast by

$$\xi_i = \Phi_i^A - \Phi_i^B \quad (4)$$

where  $\Phi_i^B$  is a similar characteristic function for constituent  $B$ . The spherical averaging of the structure factor,  $S(q)$ , is needed to eliminate the directional dependence. This ensures that only the magnitude of the scattering vector  $q$  is of relevance. This process is expressed by

$$S(q) = \sum_{|q|=|q'|} S(q') / \sum_{|q|=|q'|} 1 \quad (5)$$

where  $q = \frac{2\pi}{L}n$  and  $n$  is an integer such that  $n - \frac{1}{2} \leq \frac{qL}{2\pi}n + \frac{1}{2}$ . This encapsulates the notion that each  $n$  delineates a discrete spherical shell within which  $q$  vectors are uniformly distributed. This spherical averaging is particularly useful for systems that are isotropic, as it consolidates the information from all directions into a single,

comprehensive metric of the system structure. The structure factor is computed across discrete snapshots from the equilibrated simulation, and these individual profiles of  $S(q)$  are then methodically averaged to produce a representative measure of the morphology of the system.<sup>42</sup>

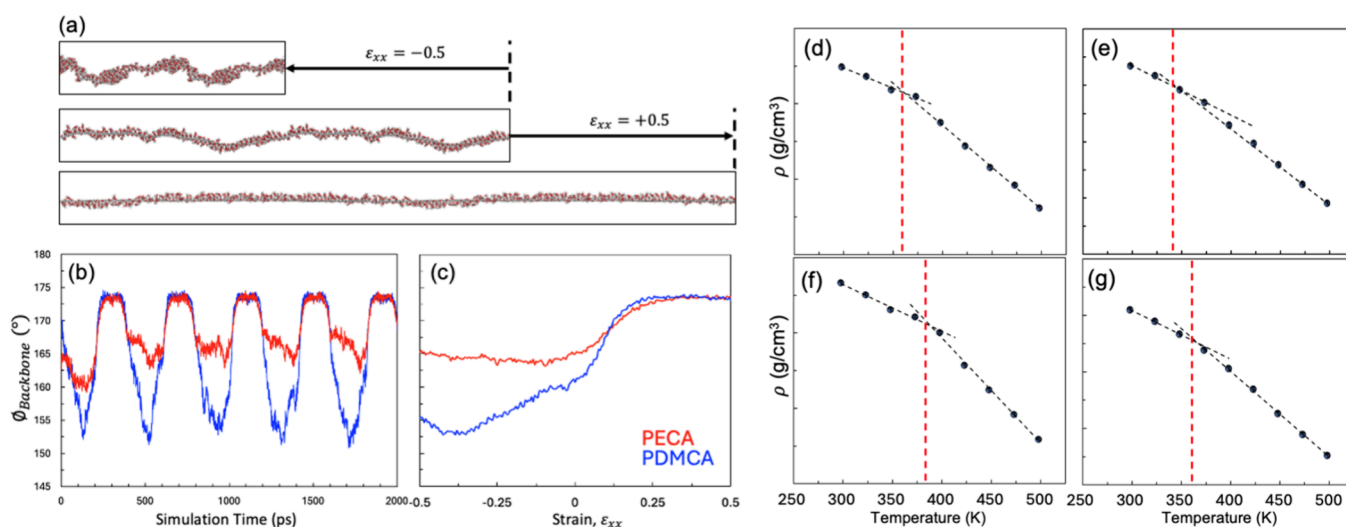
### 3. RESULTS AND DISCUSSION

#### 3.1. Simulation of the Liquid Carbonate Electrolyte.

As a baseline for comparison against their solid polymer counterparts, the solvation of  $\text{Li}^+$  ion in the liquid EC:DMC blend (1:1 molar ratio) with three levels of  $\text{LiPF}_6$  salt concentration is investigated. Particularly, the effect of salt concentration on the morphological features of the blend and ion transport is determined.

By evaluating the structure factor of the liquid systems (Figure 1a), the presence of a peak, even without salt inclusion, suggests an incomplete miscibility of EC and DMC to a certain extent. It was found that this phase segregation is enhanced more by including the  $\text{LiPF}_6$  salt at 0.5 M concentration, but then it is slightly lessened when the salt concentration is doubled. However, the change in the structure factor between





**Figure 2.** (a) Polymer undergoing tensile and compressive strains at 298 K. (b) Average backbone torsion angles as a function of simulation time under cyclic compression and stretching. (c) Corresponding backbone torsion angles plotted against applied strain for the same polymers. The change in density ( $\rho$ ) for (d) PECA, (e) PDCMA, (f) P(Et<sub>40</sub>-r-ECA<sub>60</sub>), and (g) P(Et<sub>40</sub>-r-DMCA<sub>60</sub>), as a function of temperature. The  $T_g$  is indicated by the vertical dashed line.

0.5 and 1.0 M concentrations is not significant enough to suggest a salt-mediated mechanism for mixing the solvent components. Furthermore, as depicted in Figure 1b,  $Li^+$  ion is primarily solvated by DMC, with the counterion ( $PF_6^-$ ) lying outside the first solvation shell. Further, as shown in Figure 1c–e, it is observed that DMC largely constitutes the first solvation shell surrounding  $Li^+$  ion, followed by EC and then  $PF_6^-$ , whose pair correlation peak lies  $\sim 8$  Å from  $Li^+$  ion, which suggests that  $Li^+$  and  $PF_6^-$  are fairly dissociated due to the strongly coordinating solvent molecules. We note that findings contrast with experimental ESI-MS measurements<sup>43</sup> that show that cyclic carbonates coordinate more strongly with lithium ions than linear carbonates. However, as pointed out by Seo et. al, the more volatile linear carbonates could preferentially evaporate during the ionization process and artificially diminish their apparent role in lithium coordination.<sup>43</sup>

Further, we find that  $\rho_{Li^+} - \rho_{(DMC)}$  is decreased with doubling the salt concentration, indicating the interaction between  $Li^+$  ion and the solvent molecules is slightly weakened (Figure 1f–h). However, increasing the salt concentration does not notably alter the solvation structure of  $Li^+$  ion, which is still predominantly composed of DMC, as depicted in Figure 1d,g by the coordination number of EC and DMC around  $Li^+$  ion.

Moreover, as depicted in Figure S1a, the diffusion of the  $Li^+$  ion is lower than that of the counterion ( $PF_6^-$ ) because the  $Li^+$  ion is strongly associated with solvent molecules, and thereby, its mobility is significantly suppressed compared to the  $PF_6^-$ . Despite the increase in the salt concentration having a relatively negligible effect on the solvation structure, a more pronounced effect is noticed in the diffusion profiles (Figure S1a). Namely, the diffusion of both  $Li^+$  and  $PF_6^-$  is enhanced by lower salt concentration, suggesting that a reduced ionic interaction and lower viscosity at these concentrations facilitate a greater mobility of the ions. This can be attributed to the decreased probability of ion pairing and the formation of transient network structures that typically impede ion movement at higher concentrations. Consequently, as the salt concentration decreases, the ions experience less hindrance

from these interactions, increasing their self-diffusion coefficients.

Conversely, at elevated salt concentrations, the augmented presence of ions results in enhanced ion cluster formation, which increases electrolyte viscosity and subsequently diminishes ion mobility. This effect on the viscosity is also supported by examining the MSD profiles of the solvent species at three different salt concentrations. As shown in Figure S1b, the mean square displacement (MSD) profiles of ethylene carbonate (EC) and dimethyl carbonate (DMC) exhibit distinct behaviors across different salt concentration levels. Notably, EC shows a relatively minor decrease in mobility with increasing salt concentration compared to DMC. This result aligns with the analyzed solvation structure for  $Li^+$  ions. Specifically, elevated  $Li^+$  ion concentration results in a greater number of DMC molecules solvating  $Li^+$  ions, indicating that a comparatively lesser fraction of EC molecules is involved in  $Li^+$  ion solvation.

Correspondingly, more DMC molecules are strongly tethered to  $Li^+$  ion, so their mobilities are suppressed, while EC molecules can be more mobile in the system. Please note that EC and DMC show nearly identical MSD profiles when no salt is present. This indicates that without the presence of  $Li^+$  ions, the solvents exhibit similar diffusion characteristics. As the concentration of the salt increases, the MSD for both EC and DMC decreases (with DMC undergoing a more pronounced decrease), indicating a reduction in their diffusivities. This decrease in diffusivity with increasing salt concentration establishes a key baseline for understanding ion transport in carbonate-based systems. While these liquid systems provide valuable insights, practical battery applications require solid electrolytes that combine mechanical stability with efficient ion transport. Thus, we designed polymer architectures based on the EC and DMC motifs studied above to achieve this goal.

**3.2. Solid-State Carbonate-Based Polymer Electrolytes.** **3.2.1. Polymer Architecture.** EC and DMC offer two prototypical species that serve as a platform for investigating solvents with varying polarity and viscosity.<sup>44–55</sup> Namely, an effective solid-state polymer electrolyte must offer a high

dielectric permittivity to effectively dissociate the Li salts while having sufficient mobility to facilitate ion transport. To that end, we pay attention to investigating and designing polymers based on EC and DMC. Namely, we examine poly(2-oxo-1,3-dioxolane-4-yl)methyl acrylate (PDOA), depicted in Figure S2, for our polymer electrolyte architecture due to its well-established synthesis procedure and its favorable electrochemical stability, which makes it an attractive candidate for SPE applications.<sup>6</sup> For clarity, we will refer to this polymer as PECA, standing for poly(ethylene-carbonate) acrylate.

This class of polymers was shown to have good mechanical integrity and exhibit a high  $Li^+$  ion transference number.<sup>6,7</sup> The presence of the 2-oxo-1,3-dioxolane-4-yl unit imparts the polymer with the capability to coordinate with  $Li^+$  ions and consequently facilitates the dissociation of Li salts. However, the polymer has also exhibited poor ionic conductivity at ambient temperature due to its relatively high glass transition temperature ( $>320$  K), which hinders the segmental motions necessary for ion transport. In addition, the steric hindrance imposed by the bulky side groups could limit the mobility of the  $Li^+$  ions within the polymer matrix. Therefore, the terminating 2-oxo-1,3-dioxolane-4-yl (EC) unit is replaced with DMC, yielding poly(dimethyl carbonate) acrylate (PDMCA).

**3.2.2. Mechanical Properties.** The flexibility of the polymer backbone is a critical criterion for assessing the mechanical properties and ion transport. Therefore, we analyzed the torsional characteristics of the PECA and PDMCA polymers by performing MD simulation consisting of five cycles of compression and extension at 298 K, as depicted in Figure 2a. The change of the torsional angles along the polymer backbones was monitored during the simulation (Figure 2b,c). Compared to the PDMCA polymer, the torsions in the PECA polymer are significantly restricted under compression, indicating that PECA polymer has different mechanical response compared to PDMCA, likely attributable to the bulky nature of the EC units constraining the conformational freedom of the polymer chain. During extension deformation, the torsion angles of PECA and PDMCA exhibited a comparable response, which would be largely because more space is available under the extension deformation condition. The acrylate groups in both polymers serve as pivot points, facilitating rotation and reducing steric strain, so producing a comparable tensile response. PDMCA exhibited superior overall torsional flexibility, especially in local structural modifications, owing to a reduced extent of local stacking of the DMC side groups relative to EC.

The results from the mechanical deformation simulation are consistent with the simulated glass transitions ( $T_g$ ), shown in Figure 2d,e, where PDMCA exhibits a lower  $T_g$  than PECA (see Table 1). In addition to the homopolymers (PECA and PDMCA), we also investigated random copolymers: P(Et<sub>40</sub>-r-ECA<sub>60</sub>) and P(Et<sub>40</sub>-r-DMCA<sub>60</sub>). The former is the polyacrylate with 60% of its monomeric units containing ECA in the side group, whereas the latter has 60% of its monomeric units containing DMCA in the side group.

Surprisingly, the ethylene-ECA and ethylene-DMCA copolymers exhibit higher  $T_g$  than the ECA and DMCA homopolymers. This result contradicts our expectation that these copolymers would exhibit greater mobility than the homopolymers, as we anticipated that the bulky side group would diminish the flexibility of the polymer backbone. However, we found that the lower concentration of side

**Table 1. Calculated Free Volume per Chain and  $T_g$  for a Series of Polymer Samples**

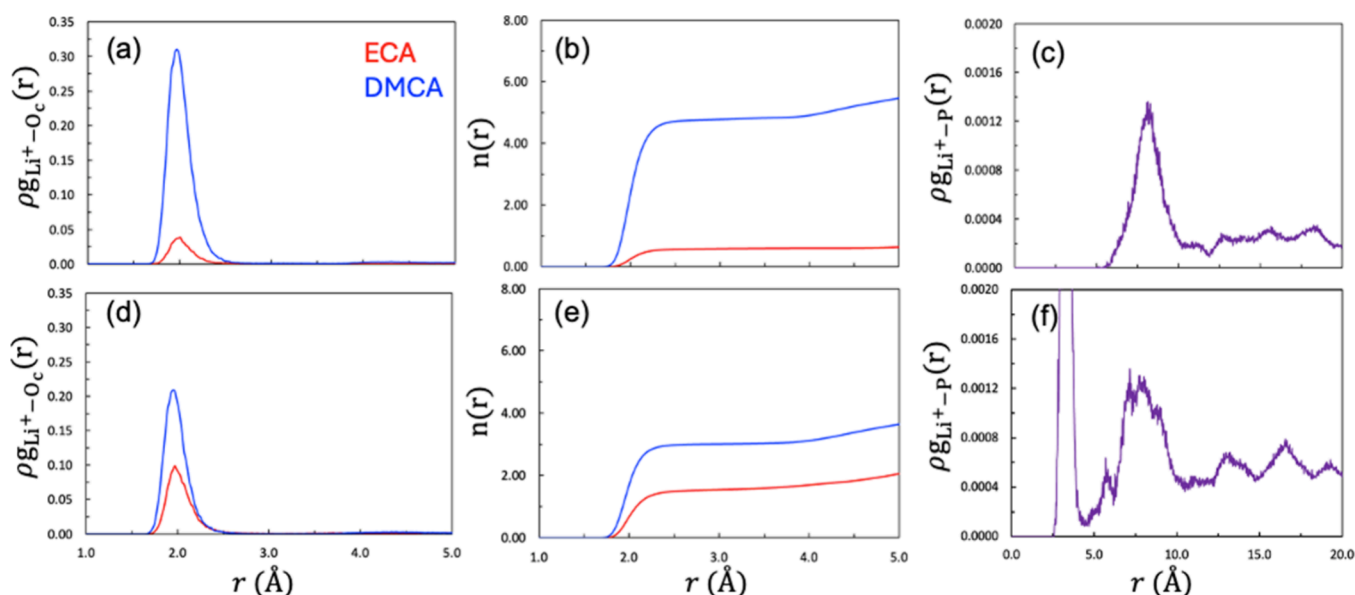
| Polymers                                                                                                           | Free Volume/Chain (Å <sup>3</sup> ) | $T_g$ (K) |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------|
| PECA                                                                                                               | 458.9                               | 349.3     |
| P(Et <sub>40</sub> -r-ECA <sub>60</sub> )                                                                          | 374.4                               | 379.0     |
| PDMCA                                                                                                              | 560.2                               | 338.5     |
| P(Et <sub>40</sub> -r-DMCA <sub>60</sub> )                                                                         | 406.4                               | 356.0     |
| P(Et <sub>40</sub> -r-ECA <sub>30</sub> -r-DMCA <sub>30</sub> )                                                    | 386.8                               | 340.2     |
| P(Et <sub>40</sub> -r-ECA <sub>60</sub> ) <sub>50</sub> + P(Et <sub>40</sub> -r-DMCA <sub>60</sub> ) <sub>50</sub> | 386.7                               | 339.9     |

groups in the copolymer chains results in smaller free volume, as summarized in Table 1, where the free volume per polymer chain with higher side group content, PECA and PDMCA, is larger compared to their counterparts with lower pendant content, P(Et<sub>40</sub>-r-ECA<sub>60</sub>) and P(Et<sub>40</sub>-r-DMCA<sub>60</sub>).

The reduced free volume is attributed to enhanced packing among the polymer chains, which in turn limits segmental motion, resulting in a higher  $T_g$ . Please note that the EC-based polymers (PECA and P(Et<sub>40</sub>-r-DMCA<sub>60</sub>)) show a distinct  $T_g$  (Figure 2d,f) due to a more abrupt change in the polymers density as they change from a glassy to a rubbery state. In contrast, the DMC-based polymers (PDMCA and P(Et<sub>40</sub>-r-DMCA<sub>60</sub>)) show a smoother transition, indicating that their chains retain greater mobility near  $T_g$ . Experimental thermal data are limited for the polymers studied here. However, a polymer synthesized from glycerol carbonate methacrylate (GCMA) has been reported with a glass transition temperature of 102 °C (375 K).<sup>56</sup> Although the specific structure and composition differ from those modeled in this work, we note that this value is close to the simulated  $T_g$  for P(Et<sub>40</sub>-r-ECA<sub>60</sub>) and related systems.

**3.2.3. Mixed-Branch Type Copolymers.** The relatively higher  $T_g$  for EC-containing chains could be further attributed to neighboring EC units, which form locally stacked structures that hinder polymer mobility. Therefore, to break such locally stacked structures, we investigate the random copolymer composed of ethylene (Et), dimethyl carbonate acrylate (DMCA), and ethylene carbonate acrylate (ECA) in a molar ratio of 40:30:30, respectively: P(Et<sub>40</sub>-r-ECA<sub>30</sub>-r-DMCA<sub>30</sub>). As shown in Table 1, the copolymer attains its glass transition at a temperature close to the  $T_g$  of the PDMCA homopolymer despite the molar concentration of DMCA units in P(Et<sub>40</sub>-r-ECA<sub>30</sub>-r-DMCA<sub>30</sub>) is only 1/3 in the PDMCA homopolymer. We find that the  $T_g$  of P(Et<sub>40</sub>-r-ECA<sub>30</sub>-r-DMCA<sub>30</sub>) is lower than other copolymers in Table 1 although the free volume of P(Et<sub>40</sub>-r-ECA<sub>30</sub>-r-DMCA<sub>30</sub>) is between that of P(Et<sub>40</sub>-r-ECA<sub>60</sub>) and P(Et<sub>40</sub>-r-DMCA<sub>60</sub>). These results imply that the comparatively low  $T_g$  of P(Et<sub>40</sub>-r-ECA<sub>30</sub>-r-DMCA<sub>30</sub>) can be attributed to other factors than the free volume.

Here, please note that the alternating arrangement of ECA and DMCA groups along the chain disrupts the regularity of the polymer chain packing and reduces the barrier required for segmental motion. In addition, its density profile as a function of temperature is also very similar to that of PDMCA and P(Et<sub>40</sub>-r-DMCA<sub>60</sub>), with a more gradual change from a glassy state to a rubbery state. The ECA, which typically confers higher  $T_g$ , might be tempered by the inclusion of DMCA, which lowers the  $T_g$  due to its less restrictive molecular structure. Consequently, the architecture of the random copolymer can be meticulously designed to optimize desirable



**Figure 3.** (a–c) RDF between  $\text{Li}^+$  ions and carbonyl oxygens in  $\text{P}(\text{Et}_{40}\text{-}r\text{-ECA}_{60})_{50} + \text{P}(\text{Et}_{40}\text{-}r\text{-DMCA}_{60})_{50}$ , local coordination numbers for carbonyl oxygens ECA and DMCA surrounding  $\text{Li}^+$  ions, and RDFs depicting  $\text{Li}^+$  ion proximity to phosphorus (P) in  $\text{PF}_6^-$  anions at a concentration of 0.5 M  $\text{LiPF}_6$ . (d–f) Corresponding RDFs and local coordination numbers at an increased concentration of 1.0 M  $\text{LiPF}_6$ . Systems are modeled at 298 K.

properties unattainable by homopolymers or single-branch-type copolymers alone.

Next, we investigated the electrolyte blend in the presence of the  $\text{Li}$  salt. As illustrated by Figure S5a, DMCA carbonyls are primarily responsible for forming the solvation shell around the  $\text{Li}^+$  ion although EC and DMC are attached on the random copolymer backbone. Such random arrangement of EC and DMC along the polymer backbone is invariant to salt concentration and temperature (as opposed to the liquid EC/DMC mixture investigated previously) and causes miscibility in 3-D space as demonstrated by the structure factor profile in Figure S5c.

**3.2.4. Blend of Single-Branch-Type Copolymers.** Although the mixed-branch copolymer approach successfully combines the properties of EC and DMC units, we investigate a more pragmatic alternative to circumvent the complex synthesis procedures often required to produce such copolymers: the physical mixing of single-branch copolymers. Therefore, we examined the 50:50 molar ratio mixture of the single-branch-type copolymers,  $\text{P}(\text{Et}_{40}\text{-}r\text{-ECA}_{60})_{50} + \text{P}(\text{Et}_{40}\text{-}r\text{-DMCA}_{60})_{50}$ .

Our simulations suggest the blend displays nearly equivalent free volume per chain as  $\text{P}(\text{Et}_{40}\text{-}r\text{-ECA}_{30}\text{-}r\text{-DMCA}_{30})$  with almost the same  $T_g$  of  $\sim 340$  K (Figure S4b and Table 1). The physical blending of these two single-branch type copolymers results in a material system with intermediate properties, effectively combining the attributes of both parent polymers without the need for complex chemical synthesis. Thus, blending two single-branch type copolymers would be potentially a more scalable (and highly tunable) approach to control polymer properties for specific applications where fine-tuning the mechanical and morphological features is necessary.

Further, this approach benefits from synergies of both parent polymers, such as the higher dielectric constant of ECA and the lower viscosity of DMCA, which could lead to an SPE with high ionic conductivity and good mechanical stability. Therefore, we investigate possible mechanisms for modulating the morphology and solvation environment in the blend (of

single-branch type copolymers) as a feasible alternative to the mixed-branch type copolymers we studied in this section.

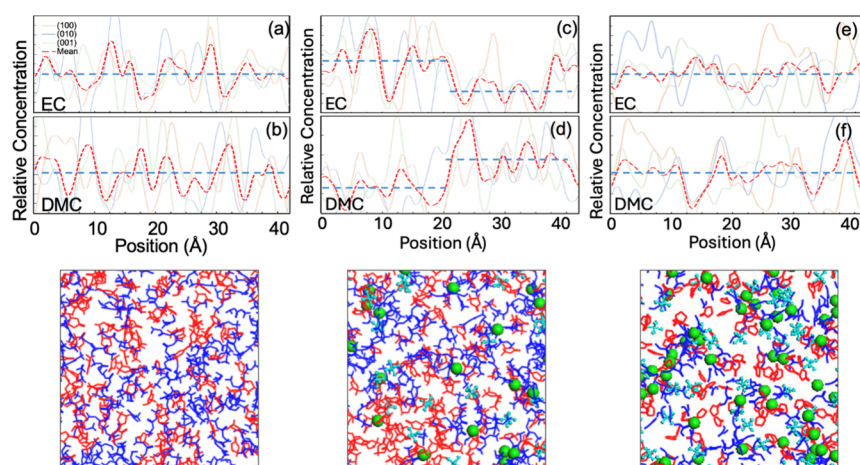
**3.2.4.1. Effect of the Salt Concentration on  $\text{Li}^+$  Solvation.** Understanding how morphology and solvation environment evolve in the polymer blend provides insight into its viability as an alternative to the mixed-branch copolymers studied earlier. To explore this, we examine how  $\text{Li}$  salt concentration influences both local solvation and larger-scale morphology. Namely, we study three salt-to-solvating carbonyl oxygen ratios ( $\text{Li}:\text{C}=\text{O}$ ): 0, 0.04, and 0.09, approximately corresponding to 0.0, 0.5, and 1.0 M  $\text{LiPF}_6$  concentrations.

As shown in Figure 3a–c, the relative distribution and coordination of the carbonate species and the counterion at 0.5 M  $\text{LiPF}_6$  closely resemble those of the liquid electrolyte at a comparable concentration. In both states, DMC (or DMCA) carbonyl oxygen substantially dominates the first solvation shell around the  $\text{Li}^+$  ion.

As mentioned previously, the solvation of  $\text{Li}^+$  presents a point of contention in the field, with some simulations indicating a coordination favoring EC due to its higher dielectric constant,<sup>46,49,51</sup> while other works suggest either equivalent coordination or DMC–dominant coordination.<sup>43,44,52,54,55</sup> That said, although our nonbonded force field parameters were meticulously fitted to DFT calculations, we acknowledge that using Mulliken charges may underestimate the polarity of EC and potentially overstate the entropic contributions of DMC relative to the strong interactions of EC with  $\text{Li}^+$ . Nevertheless, we anticipate that the observed patterns in morphological behavior across varying salt concentrations and temperatures remain qualitatively sound, as the underlying physical mechanisms would continue to function irrespective of the precise EC/DMC solvation proportions.

To directly investigate this DMC dominance phenomenon, we conducted a series of simulations at 298 K over 10 ns, modeling the  $\text{Li}^+$  ion coordinated with four solvent molecules in a vacuum while systematically varying the relative proportions of EC and DMC molecules. These simulations





**Figure 4.** Relative concentration profiles of ECA and DMCA along the 100, 010, and 001 directions in the simulation box (with the red color denoting the mean of all three directions) for (a, b) 0.0 M  $\text{LiPF}_6$ , (c, d) 0.5 M  $\text{LiPF}_6$ , and (e, f) 1.0 M  $\text{LiPF}_6$ .

revealed that the liquid and solid electrolyte environments differ significantly from the vacuum conditions (Figure S6), likely due to the greater conformational versatility of DMC, accommodating the spatial and orientational requirements of the surrounding ions and molecules in the bulk phase. Since DMC dominates the  $\text{Li}^+$  ion solvation despite the weak interaction with the  $\text{Li}^+$  ion, it is inferred that the superior conformational entropy of DMC is likely the driving factor for the preference of DMC over EC within the bulk phase of the electrolyte. This result is in line with past studies that determined that the dipole moment of DMC is substantially amplified upon coordination with  $\text{Li}^+$  due to undergoing a conformational change.<sup>43</sup>

Interestingly, unlike the liquid,  $\text{Li}^+$  ion solvation is substantially altered when the salt concentration is nearly doubled. Namely, as shown in Figure 3d,e, ECA now plays a much more equitable role in  $\text{Li}^+$  ion solvation compared to DMCA. Further, as evidenced by Figure 3f, even the counterion ( $\text{PF}_6^-$ ) interacts more closely with  $\text{Li}^+$  ion than the lower salt concentration in the polymer blend and both the low and high salt concentrations in the liquid blend. In the case of liquid electrolytes, increasing the salt concentration did not substantially alter the solvation preference due to the high configurational entropy facilitated by DMC, as shown previously. This configurational freedom allows the coordination to maintain its preference for DMC even when the enthalpic interactions would favor EC. Conversely, in solid polymer electrolytes, the polymer backbone introduces spatial constraints that can restrict the conformational freedom of the solvent molecules, including both EC and DMC. When salt concentration increases in a polymer blend, the solvent molecules are less capable of maneuvering and maintaining their preferred conformations. The EC (with its stronger enthalpic interactions with  $\text{Li}^+$  ions) exhibits increased competitiveness in solvation within this constrained environment, as its enthalpic advantages are not counterbalanced by entropic factors, unlike in the liquid state.

**3.2.4.2. Effect of Temperature on  $\text{Li}^+$  Solvation.** Although salt concentration strongly influences the properties of polymer blends, practical battery applications require an understanding of how these systems behave across a range of temperatures. We systematically examine temperature effects on both molecular-scale solvation and nanoscale morphology.

At 0.5 M salt concentration, increasing the temperature from 298 to 448 K significantly reduces the number of coordinating ECA around  $\text{Li}^+$  ions, while DMCA remains relatively stable (see Figure S8). When the temperature reaches 448 K, which exceeds the glass transition temperature of the polymer, ECA barely coordinates with  $\text{Li}^+$  ions. Specifically, the coordination number of ECA diminishes, whereas that of DMCA slightly increases from 4.8 to 5.2, as seen in Figure S8b,e,h.

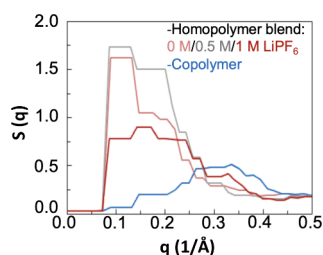
On the other hand, at higher salt concentrations (1.0M), the system behaves differently. As seen in Figure S12, ECA maintains stronger coordination with  $\text{Li}^+$  ions even at elevated temperatures, with its coordination number decreasing from 1.5 to 1.2 between 298 and 448 K. Additionally, as temperature increases,  $\text{PF}_6^-$  ions show reduced coordination with  $\text{Li}^+$  ions, suggesting enhanced salt dissociation, which contrasts with the 0.5 M concentration, where  $\text{PF}_6^-$  coordination stays consistently low across all temperatures (Figure S8). Temperature affects the molecular competition between enthalpic and entropic factors differently at various salt concentrations. Namely, the sustained presence of ECA around lithium ions at 1.0M, even above the glass transition temperature, warrants further investigation.

**3.2.4.3. Effect of the Salt Concentration on Nanophase Morphology.** We extend our inquiry to examine how the inclusion of Li salts can further influence the morphology and properties of the blend. Modulating the Li salt content is a well-known strategy to tune ionic conductivity in polymer electrolytes.<sup>15</sup> However, their influence on the morphology and phase behavior of polymer electrolytes has not been well elucidated.

The introduction of salt at 0.5 M concentration increases the phase separation between the ECA and DMCA-based polymers, as shown in the concentration profiles of Figure 4. Interestingly, doubling the salt concentration results in a reduction of this phase separation. The increased presence of the salt likely disrupts the microphase separation by interacting with both ECA and DMCA branches, which leads to a more homogeneous polymer matrix. As the concentration of salt increases, the ions can facilitate bridging interactions between the different segments, reducing the phase separation and creating a more uniform electrolyte environment.

The morphological tendency can be more clearly explained by computing the structural factor between ECA and DMCA

for various blends and polymer systems, as shown in Figure 5. The structure factor profile ( $S(q)$ ) for the blend without added



**Figure 5.** Structure factor between ECA and DMCA for various systems. Here, *blend* denotes  $P(\text{Et}_{40}\text{-}r\text{-ECA}_{60})_{50} + P(\text{Et}_{40}\text{-}r\text{-DMCA}_{60})_{50}$  (with the three salt concentrations), while *copolymer* denotes  $P(\text{Et}_{40}\text{-}r\text{-ECA}_{30}\text{-}r\text{-DMCA}_{30})$ .

salt shows distinct peaks indicative of significant phase separation between ECA and DMCA domains. When 0.5 M  $\text{LiPF}_6$  is introduced, these peaks become more pronounced, which suggests enhanced phase separation. This result is likely due to the preferential interaction of  $\text{Li}^+$  ions with  $P(\text{Et}_{40}\text{-}r\text{-DMCA}_{60})$  over  $P(\text{Et}_{40}\text{-}r\text{-ECA}_{60})$ . However, doubling the salt concentration to 1 M results in a starkly different  $S(q)$  profile, with the previously sharp peaks now significantly reduced, which is a strong indication of reduced phase separation. Therefore,  $\text{Li}^+$  ions effectively act as a compatibilizer at higher salt concentrations by facilitating interactions between the polymer components, which leads to improved miscibility. This aligns well with the previously observed radial distributions of ECA and DMCA surrounding  $\text{Li}^+$  ions at higher salt concentrations. The ability of the salt content to effectively modulate the nanophase morphology provides a method to tailor the mechanical and conductive properties of SPEs without chemically modifying the polymer structure.

**3.2.4.4. Effect of Temperature on Nanophase Morphology.** Here, we focus on the influence of temperature on the nanophase morphology of the polymer blends. As depicted in Figure 6a, the morphology is not significantly affected at temperatures below or not substantially above the  $T_g$  of  $\sim 340$  K (Table 2). This is because molecular mobility is still relatively constrained within this temperature range, and no significant polymer chain rearrangement can occur until the polymer network transitions into the rubbery state, where increased segmental motion allows for larger-scale morphological alterations. Only when the polymer is heated to 423 K, a significant decrease of the  $S(q)$  peak is observed. Indeed, the peak is substantially diminished at 498 K.

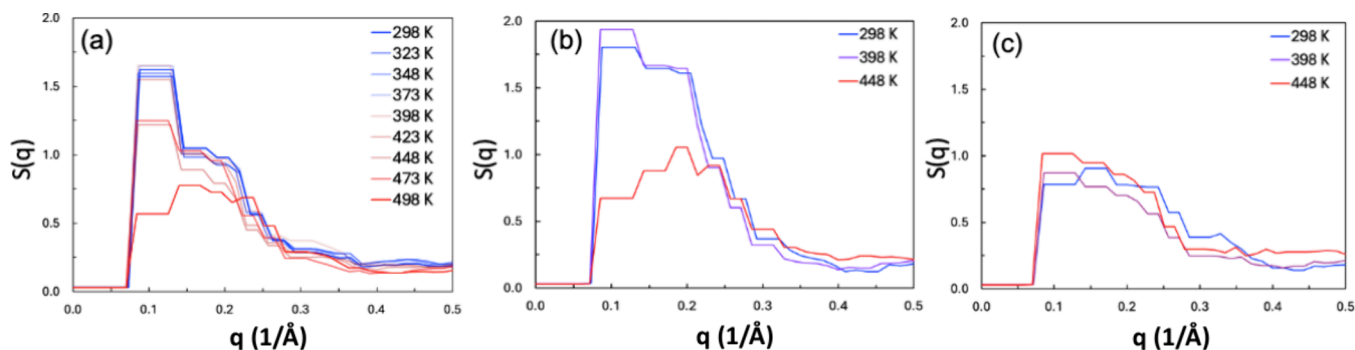
**Table 2.** Coordination Numbers of the Carbonyl Oxygens Belonging to the Different Solvent Species around  $\text{Li}^+$  Ion

| System | Temperature (K) | Salt Concent. (M) | EC/ECA CN | DMC/DMCA CN |
|--------|-----------------|-------------------|-----------|-------------|
| SPE    | 298             | 0.5               | 0.600     | 4.836       |
|        | 398             | 0.5               | 0.440     | 4.908       |
|        | 448             | 0.5               | 0.190     | 5.173       |
|        | 298             | 1.0               | 1.543     | 3.016       |
|        | 398             | 1.0               | 1.563     | 3.621       |
|        | 448             | 1.0               | 1.220     | 4.000       |
| Liquid | 298             | 0.5               | 0.579     | 5.084       |
|        | 298             | 1.0               | 0.676     | 5.259       |

Then, in Figure 6b, a similar trend is observed for 0.5 M concentration, where the peak is not significantly altered until temperatures are well above the  $T_g$ . This might be due to the more gradual shift from the glassy to the rubbery state of this polymer blend, indicating that the polymers have not fully transitioned into the rubbery state (even at temperatures slightly above the  $T_g$ ). Another reason could be the persistence of some form of local ordering or physical cross-links, which prevents the full realization of the rubbery state of the polymer at temperatures even above but close to  $T_g$ .

On the other hand, as shown in Figure 6c for the 1.0 M concentration, the structure factor  $S(q)$  peaks are not altered, even at 448 K, indicating that the miscibility is enhanced at high salt concentration regardless of temperature change. This is evidenced by this  $S(q)$  peak intensity being similar across all the salt concentrations at the elevated temperature (498 K). At the 1.0 M concentration,  $\text{Li}^+$  ions disrupt the nascent phase boundaries, which induces a more uniform and stable morphology less sensitive to temperature-induced rearrangements. This could potentially have significant implications for the performance of SPEs. Namely, this suggests that high-salt-concentration polymer blends could maintain a stable morphology over a broad temperature range.

However, while structural stability is important, practical performance also hinges on ion transport. Our analysis of  $\text{Li}^+$  ion mobility reveals striking differences between low and high salt concentrations at 0.5 and 1.0 M (Figures S9–S11). At 398 K, both concentrations exhibit subdiffusive behavior, characterized by  $\text{MSD} \propto t^\alpha$  with  $\alpha < 1$ , indicating constrained ion diffusion. In the 1.0 M system,  $\text{Li}^+$  remains effectively trapped with MSD below  $1 \text{ \AA}^2$  over 400 ns, whereas in the 0.5 M system the MSD gradually approaches a linear time dependence upon heating. Above  $T_g$  (448 K), the 0.5 M system achieves Fickian diffusion ( $\alpha \rightarrow 1$ ) with diffusion coefficients of



**Figure 6.** Structure factor between ECA and DMCA for different temperatures for (a) 0.0 M  $\text{LiPF}_6$ , (b) 0.5  $\text{LiPF}_6$ , and (c) 1.0 M  $\text{LiPF}_6$ .



$\sim 0.97 \times 10^{-9}$  and  $3.27 \times 10^{-9}$  cm<sup>2</sup>/s for Li<sup>+</sup> and PF<sub>6</sub><sup>−</sup> ions, respectively, while the 1.0 M system remains subdiffusive. The pronounced disparity arises from the less constrained solvation shell at 0.5 M concentration, predominantly consisting of DMCA, in contrast to the enduring ECA coordination at 1.0 M concentration, which forms an enthalpic cage around the lithium ions due to the robust interaction between Li<sup>+</sup> and the carbonyl group of ECA.

Given that the system with 0.5 M concentration achieved Fickian diffusion only at 448 K (well above typical battery operating temperatures), reducing the glass transition temperature becomes crucial for achieving practical ionic conductivity. One promising approach is to incorporate plasticizing solvent molecules. For example, adding 25 wt % of either EC or DMC to P(Et<sub>40</sub>-*r*-ECA<sub>30</sub>-*r*-DMCA<sub>30</sub>) lowers the glass transition temperature to 317.3 K with EC and even further to 278.7 K with DMC, compared to 340 K for the neat polymer (Figure S13). Structure factor analysis of these blends reveals excellent miscibility even in the absence of lithium salt, offering a promising pathway for optimizing polymer electrolyte performance. Future studies should assess how such plasticizers influence salt concentration-dependent behavior and whether they can preserve the morphological stability observed at higher salt concentrations while also enhancing ion mobility.

#### 4. SUMMARY

This study employs all-atom molecular dynamics simulations to explore Li<sup>+</sup> ion solvation, diffusion, and morphology in carbonate-based electrolytes, particularly focusing on ethylene carbonate (EC) and dimethyl carbonate (DMC) in both liquid and solid polymeric forms. The liquid electrolyte phase reveals a dominance of DMC in Li<sup>+</sup> solvation at low salt concentrations, while EC becomes competitive at higher concentrations. Interestingly, despite experimental claims that EC coordinates more strongly with Li<sup>+</sup> ion, the simulations attribute the preference of DMC to its greater conformational entropy in bulk environments.

In the solid phase, polymers incorporating DMC (like PDMCA) exhibit lower glass transition temperatures and greater chain flexibility than their EC-based counterparts (like PECA), translating to improved ion mobility. The study further evaluates copolymers and physically blended polymer systems, which combine the dielectric strength of EC with the structural softness of DMC. These mixed systems can be optimized to enhance mechanical robustness and ionic conductivity without requiring complex synthesis techniques.

The presence of Li salt has a concentration-dependent impact: at moderate concentrations, it induces phase separation, while at higher concentrations, it acts as a compatibilizer, improving miscibility between EC and DMC domains. Temperature further modulates these behaviors—especially above the glass transition, where entropic effects reduce coordination of EC to Li<sup>+</sup> ion unless the salt concentrations are high.

In high-salt systems, ion mobility becomes severely restricted due to enthalpic traps formed by EC, whereas low-salt blends favor diffusion, though only at elevated temperatures. To resolve this, adding plasticizers like free DMC or EC significantly reduces the glass transition temperature and preserves miscibility, offering a practical pathway to optimize solid-state polymer electrolytes.

Several predictions made in this work—including glass transition behavior, ion mobility, and phase morphology can be evaluated experimentally using DSC, PFG-NMR, impedance spectroscopy, or SAXS/SANS. These techniques offer a path to validate the simulated trends in segmental dynamics, diffusion, and nanostructure. Additionally, the simulation framework presented here can serve as a validated baseline for future refinement through AIMD studies or polarizable force field models that better capture electrostatics under varied salt and temperature conditions.

Overall, the findings emphasize that Li<sup>+</sup> ion transport in carbonate-based polymer electrolytes is a delicate balance of solvation dynamics, salt concentration, morphology, and temperature. These molecular insights establish a foundation for tailoring next-generation solid polymer electrolytes for safe, high-performance lithium batteries.

#### ■ ASSOCIATED CONTENT

##### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.chemmater.5c01016>.

Mean-square displacement of ions, chemical structures of polymers, density change with increasing temperature, radial distribution functions between Li ions and neighboring oxygen atoms, energy curves, force field parameters, and simulation box dimensions (PDF)

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##### Notes

The authors declare no competing financial interest.

#### ■ REFERENCES

- (1) Larcher, D.; Tarascon, J. M. Towards greener and more sustainable batteries for electrical energy storage. *Nat. Chem.* **2015**, *7* (1), 19–29.
- (2) He, Q.; Yu, B.; Li, Z. H.; Zhao, Y. Density Functional Theory for Battery Materials. *Energy Environ. Mater.* **2019**, *2* (4), 264–279.
- (3) Meng, Y. S.; Arroyo-de Dompablo, M. E. First principles computational materials design for energy storage materials in lithium ion batteries. *Energy Environ. Sci.* **2009**, *2* (6), 589–609.
- (4) Nitta, N.; Wu, F. X.; Lee, J. T.; Yushin, G. Li-ion battery materials: present and future. *Mater. Today* **2015**, *18* (5), 252–264.
- (5) Bruce, P. G.; Scrosati, B.; Tarascon, J. M. Nanomaterials for rechargeable lithium batteries. *Angew. Chem., Int. Ed.* **2008**, *47* (16), 2930–2946.
- (6) Mindemark, J.; Lacey, M. J.; Bowden, T.; Brandell, D. Beyond PEO-Alternative host materials for Li-conducting solid polymer electrolytes. *Prog. Polym. Sci.* **2018**, *81*, 114–143.
- (7) Zhao, Y. B.; Bai, Y.; Li, W. D.; An, M. Z.; Bai, Y. P.; Chen, G. R. Design Strategies for Polymer Electrolytes with Ether and Carbonate

Groups for Solid-State Lithium Metal Batteries. *Chem. Mater.* **2020**, *32* (16), 6811–6830.

(8) Hassoun, J.; Panero, S.; Reale, P.; Scrosati, B. A New, Safe, High-Rate and High-Energy Polymer Lithium-Ion Battery. *Adv. Mater.* **2009**, *21* (47), 4807–4810.

(9) Xue, Z. G.; He, D.; Xie, X. L. Poly(ethylene oxide)-based electrolytes for lithium-ion batteries. *J. Mater. Chem. A* **2015**, *3* (38), 19218–19253.

(10) Soliman, W. G.; Swathi, C.; Yasasvi, T.; Priya, B. K.; Reddy, D. A. Review on poly(ethylene oxide)-based electrolyte and anode nanomaterials for the internet of things node-level lithium-ion batteries. *Mater. Today: Proc.* **2021**, *42*, 429–435.

(11) Faliya, K.; Kliem, H. Charge distributions in poly(ethylene oxide)-based electrolytes for lithium-ion batteries. *Electrochim. Acta* **2018**, *290*, 211–219.

(12) Lee, J.; Kim, B. S. Recent progress in poly(ethylene oxide)-based solid-state electrolytes for lithium-ion batteries. *Bull. Korean Chem. Soc.* **2023**, *44* (10), 831–840.

(13) Homann, G.; Stolz, L.; Nair, J.; Laskovic, I. C.; Winter, M.; Kasnatscheew, J. Poly(Ethylene Oxide)-based Electrolyte for Solid-State-Lithium-Batteries with High Voltage Positive Electrodes: Evaluating the Role of Electrolyte Oxidation in Rapid Cell Failure. *Sci. Rep.* **2020**, *10* (1), 4390.

(14) Brooks, D. J.; Merinov, B. V.; Goddard, W. A.; Kozinsky, B.; Mailoa, J. Atomistic Description of Ionic Diffusion in PEO-LiTFSI: Effect of Temperature, Molecular Weight, and Ionic Concentration. *Macromolecules* **2018**, *51* (21), 8987–8995.

(15) Molinari, N.; Mailoa, J. P.; Kozinsky, B. Effect of Salt Concentration on Ion Clustering and Transport in Polymer Solid Electrolytes: A Molecular Dynamics Study of PEO-LiTFSI. *Chem. Mater.* **2018**, *30* (18), 6298–6306.

(16) Ennari, J.; Pietilä, L. O.; Virkkunen, V.; Sundholm, F. Molecular dynamics simulation of the structure of an ion-conducting PEO-based solid polymer electrolyte. *Polymer* **2002**, *43* (20), 5427–5438.

(17) Polaskova, M.; Peer, P.; Cermak, R.; Ponizil, P. Effect of Thermal Treatment on Crystallinity of Poly(ethylene oxide) Electrospun Fibers. *Polymers* **2019**, *11* (9), 1384.

(18) Yang, H. C.; Zhang, Y. M.; Tennenbaum, M. J.; Althouse, Z.; Ma, Y.; He, Y. B.; Wu, Y. T.; Wu, T. H.; Mathur, A.; Chen, P.; et al. Polypropylene Carbonate-Based Adaptive Buffer Layer for Stable Interfaces of Solid Polymer Lithium Metal Batteries. *ACS Appl. Mater. Interfaces* **2019**, *11* (31), 27906–27912.

(19) He, Y. B.; Liu, N. A.; Kohl, P. A. High Conductivity, Lithium Ion Conducting Polymer Electrolyte Based on Hydrocarbon Backbone with Pendent Carbonate (vol 167, 100517, 2020). *J. Electrochem. Soc.* **2020**, *167* (10), No. 109002.

(20) Didwal, P. N.; Singhbabu, Y. N.; Verma, R.; Sung, B. J.; Lee, G. H.; Lee, J. S.; Chang, D. R.; Park, C. J. An advanced solid polymer electrolyte composed of poly(propylene carbonate) and mesoporous silica nanoparticles for use in all-solid-state lithium-ion batteries. *Energy Storage Mater.* **2021**, *37*, 476–490.

(21) Bouchet, R.; Maria, S.; Meziane, R.; Aboulaich, A.; Lienafa, L.; Bonnet, J. P.; Phan, T. N. T.; Bertin, D.; Gigmes, D.; Devaux, D.; et al. Single-ion BAB triblock copolymers as highly efficient electrolytes for lithium-metal batteries. *Nat. Mater.* **2013**, *12* (5), 452–457.

(22) Ji, J. Y.; Keen, J.; Zhong, W. H. Simultaneous improvement in ionic conductivity and mechanical properties of multi-functional block-copolymer modified solid polymer electrolytes for lithium ion batteries. *J. Power Sources* **2011**, *196* (23), 10163–10168.

(23) Zittlau, P.; Mross, S.; Gond, D.; Kohns, M. Molecular modeling and simulation of organic electrolyte solutions for lithium ion batteries. *J. Chem. Phys.* **2024**, *161* (12), 12.

(24) Muralidharan, A.; Chaudhari, M. I.; Pratt, L. R.; Rempe, S. B. Molecular Dynamics of Lithium Ion Transport in a Model Solid Electrolyte Interphase. *Sci. Rep.* **2018**, *8*, 8.

(25) Thompson, A. P.; Aktulga, H. M.; Berger, R.; Bolintineanu, D. S.; Brown, W. M.; Crozier, P. S.; Veld, P. J. I.; Kohlmeyer, A.; Moore, S. G.; Nguyen, T. D.; et al. LAMMPS—a flexible simulation tool for

particle-based materials modeling at the atomic, meso, and continuum scales. *Comput. Phys. Commun.* **2022**, *271*, No. 108171.

(26) Mayo, S. L.; Olafson, B. D.; Goddard, W. A. Dreiding - a Generic Force-Field for Molecular Simulations. *J. Phys. Chem.* **1990**, *94* (26), 8897–8909.

(27) Jensen, K. P.; Jorgensen, W. L. Halide, Ammonium, and Alkali Metal Ion Parameters for Modeling Aqueous Solutions. *J. Chem. Theory Comput.* **2006**, *2* (6), 1499–1509.

(28) Delley, B. From molecules to solids with the DMol approach. *J. Chem. Phys.* **2000**, *113* (18), 7756–7764.

(29) Delley, B. An All-Electron Numerical-Method for Solving the Local Density Functional for Polyatomic-Molecules. *J. Chem. Phys.* **1990**, *92* (1), 508–517.

(30) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **1996**, *77* (18), 3865–3868.

(31) Perdew, J. P.; Burke, K.; Wang, Y. Generalized gradient approximation for the exchange-correlation hole of a many-electron system. *Phys. Rev. B* **1996**, *54* (23), 16533–16539.

(32) Wilson, A. K.; vanMourik, T.; Dunning, T. H. Gaussian basis sets for use in correlated molecular calculations 0.6. Sextuple zeta correlation consistent basis sets for boron through neon. *J. Mol. Struct. THEOCHEM* **1996**, *388*, 339–349.

(33) Van Mourik, T.; Dunning, T. H. Gaussian basis sets for use in correlated molecular calculations. VIII. Standard and augmented sextuple zeta correlation consistent basis sets for aluminum through argon. *Int. J. Quantum Chem.* **2000**, *76* (2), 205–221.

(34) Prascher, B. P.; Woon, D. E.; Peterson, K. A.; Dunning, T. H.; Wilson, A. K. Gaussian basis sets for use in correlated molecular calculations. VII. Valence, core-valence, and scalar relativistic basis sets for Li, Be, Na, and Mg. *Theor. Chem. Acc.* **2011**, *128* (1), 69–82.

(35) Hill, J. G.; Peterson, K. A. Gaussian basis sets for use in correlated molecular calculations. XI. Pseudopotential-based and all-electron relativistic basis sets for alkali metal (K-Fr) and alkaline earth (Ca-Ra) elements. *J. Chem. Phys.* **2017**, *147* (24), 244106.

(36) Dunning, T. H.; Peterson, K. A.; Wilson, A. K. Gaussian basis sets for use in correlated molecular calculations. X. The atoms aluminum through argon revisited. *J. Chem. Phys.* **2001**, *114* (21), 9244–9253.

(37) Dunning, T. H. Gaussian-Basis Sets for Use in Correlated Molecular Calculations 0.1. The Atoms Boron through Neon and Hydrogen. *J. Chem. Phys.* **1989**, *90* (2), 1007–1023.

(38) Mulliken, R. S. Electronic Population Analysis on Lcao-Mo Molecular Wave Functions 0.3. Effects of Hybridization on Overlap and Gross Ao Populations. *J. Chem. Phys.* **1955**, *23* (12), 2338–2342.

(39) Mulliken, R. S.; et al. *J. Chem. Phys.* **1955**, *23* (10), 1841–1846.

(40) Mulliken, R. S. Electronic Population Analysis on Lcao-Mo Molecular Wave Functions 0.4. Bonding and Antibonding in Lcao and Valence-Bond Theories. *J. Chem. Phys.* **1955**, *23* (12), 2343–2346.

(41) Mulliken, R. S. Electronic Population Analysis on Lcao-Mo Molecular Wave Functions 0.1. *J. Chem. Phys.* **1955**, *23* (10), 1833–1840.

(42) Allam, O. *Atomistic Modeling and Machine Learning for the Rational Design of Organic Energy Storage Materials*. Georgia Institute of Technology, 2024.

(43) Seo, D. M.; Reining, S.; Kutcher, M.; Redmond, K.; Euler, W. B.; Lucht, B. L. Role of Mixed Solvation and Ion Pairing in the Solution Structure of Lithium Ion Battery Electrolytes. *J. Phys. Chem. C* **2015**, *119* (25), 14038–14046.

(44) Haghkhah, H.; Chooobar, B. G.; Amjad-Iranagh, S. Effect of salt concentration on properties of mixed carbonate-based electrolyte for Li-ion batteries: a molecular dynamics simulation study. *J. Mol. Model.* **2020**, *26* (8), 220.

(45) Boyer, M. J.; Vilciauskas, L.; Hwang, G. S. Structure and Li ion transport in a mixed carbonate/LiPF<sub>6</sub> electrolyte near graphite electrode surfaces: a molecular dynamics study. *Phys. Chem. Chem. Phys.* **2016**, *18* (40), 27868–27876.

(46) Liu, M.; Chintali, P. J.; Huang, X. B.; Zhang, R. B. Structures and dynamic properties of the LiPF<sub>6</sub> electrolytic solution under

electric fields - a theoretical study. *Phys. Chem. Chem. Phys.* **2019**, *21* (24), 13186–13193.

(47) You, X. L.; Chaudhari, M. I.; Rempe, S. B.; Pratt, L. R. Dielectric Relaxation of Ethylene Carbonate and Propylene Carbonate from Molecular Dynamics Simulations. *J. Phys. Chem. B* **2016**, *120* (8), 1849–1853.

(48) Han, S. H. Structure and dynamics in the lithium solvation shell of nonaqueous electrolytes. *Sci. Rep.* **2019**, *9*, 5555.

(49) Han, S. A salient effect of density on the dynamics of nonaqueous electrolytes. *Sci. Rep.* **2017**, *7*, 46718.

(50) Tenney, C. M.; Cygan, R. T. Analysis of Molecular Clusters in Simulations of Lithium-Ion Battery Electrolytes. *J. Phys. Chem. C* **2013**, *117* (47), 24673–24684.

(51) Postupna, O. O.; Kolesnik, Y. V.; Kalugin, O. N.; Prezhdo, O. V. Microscopic Structure and Dynamics of  $\text{LiBF}_4$  Solutions in Cyclic and Linear Carbonates. *J. Phys. Chem. B* **2011**, *115* (49), 14563–14571.

(52) Soetens, J. C.; Millot, C.; Maigret, B. Molecular dynamics simulation of  $\text{Li}^+\text{BF}_4^-$  in ethylene carbonate, propylene carbonate, and dimethyl carbonate solvents. *J. Phys. Chem. A* **1998**, *102* (7), 1055–1061.

(53) Masia, M.; Probst, M.; Rey, R. Ethylene carbonate- $\text{Li}^+$ : A theoretical study of structural and vibrational properties in gas and liquid phases. *J. Phys. Chem. B* **2004**, *108* (6), 2016–2027.

(54) Skarmoutsos, I.; Ponnuchamy, V.; Vetere, V.; Mossa, S.  $\text{Li}^+$  Solvation in Pure, Binary, and Ternary Mixtures of Organic Carbonate Electrolytes. *J. Phys. Chem. C* **2015**, *119* (9), 4502–4515.

(55) Borodin, O.; Smith, G. D. Quantum Chemistry and Molecular Dynamics Simulation Study of Dimethyl Carbonate: Ethylene Carbonate Electrolytes Doped with  $\text{LiPF}_6$ . *J. Phys. Chem. B* **2009**, *113* (6), 1763–1776.

(56) Büttner, H.; Kohrt, C.; Wultia, C.; Schäffner, B.; Groenke, K.; Hu, Y. Y.; Kruse, D.; Werner, T. Life Cycle Assessment for the Organocatalytic Synthesis of Glycerol Carbonate Methacrylate. *ChemSusChem* **2019**, *12* (12), 2701–2707.



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